

Answer Key — Why Earth Is Unique: Its Position

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **right distance from the Sun** — the right distance keeps water liquid.
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- Q2** (b) **habitable zone** — that range is the habitable (Goldilocks) zone.
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- Q3** (b) **Venus** — Venus is hottest due to its thick CO₂ atmosphere.
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- Q4** (b) **about 70% of its surface is water** — vast water cover makes Earth look blue from space.
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- Q5** (a) **Both true, R explains A** — the trapped heat (greenhouse effect) makes Venus hottest — as R explains.
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- Q6** **greenhouse** — this heat-trapping is the greenhouse effect.
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- Q7** **liquid** — liquid water is essential for life.
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Short answer

- Q8** It is just right — not so close that water evaporates nor so far that it freezes — so water stays liquid, which is essential for life.
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- Q9** The range of distances from a star over which water can remain in the liquid state; also called the Goldilocks zone.
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- Q10** Gases like carbon dioxide trap some of the Sun's heat, keeping Earth's temperature just right for liquid water; without it Earth would be too cold.
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Long / application

- Q11** Earth is at the right distance from the Sun and has a mild greenhouse effect, keeping temperatures suitable for liquid water and life. Venus, though farther from the Sun than Mercury, has a thick carbon dioxide atmosphere that traps heat strongly (runaway greenhouse effect), making it the hottest planet — far too hot for liquid water or life.
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- Q12** Probably not, because beyond the habitable zone it is too cold — the water would freeze and there would not be the right temperature for plant life processes. Liquid water and a suitable temperature are essential, which such a planet lacks.
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